

CASE 4 — JPMorgan: From Blockchain Experiment to Programmable Money

Company: J.P. Morgan

Blockchain experimentation: 2015–2016 onward

Quorum: 2016

JPM Coin: 2019

Current platform: Kinexys

Sector: Banking and financial infrastructure

The story

Students often associate blockchain with the idea of removing banks.

This case turns that assumption upside down:

What happens when one of the world's major banks decides to use blockchain itself?

J.P. Morgan established its Blockchain Center of Excellence in **2015** and launched **Quorum**, a permissioned blockchain designed for financial institutions and enterprises, in **2016**.

In **2017**, it worked with the Monetary Authority of Singapore on blockchain-based multi-bank payment experiments under Project Ubin.

In **2019**, it launched JPM Coin, now part of what is called **Kinexys Digital Payments**.

The important engineering question is therefore no longer simply:

"Can blockchain transfer Bitcoin?"

It becomes:

"Can distributed-ledger infrastructure improve institutional payments, settlement and tokenized assets?"

J.P. Morgan reports that its blockchain infrastructure subsequently expanded into digital payments and tokenized assets; its materials state that its digital-assets platform has settled more than **\$900 billion in tokenized U.S. Treasuries**.

The traditional problem

Suppose:

Company A → Bank A → correspondent bank → Bank B → Company B

Payments and settlement can involve multiple intermediaries, databases, reconciliation processes and operating-hour constraints.

A blockchain-based institutional system explores a different architecture:

Company A → shared programmable infrastructure → Company B

subject, of course, to banking controls, permissions and regulation.

Focus

Blockchain as financial infrastructure and programmable settlement.

Students should investigate:

tokenization – permissioned blockchain – settlement – smart contracts – programmable payments

The engineering issue

Why use blockchain at all?

If J.P. Morgan already possesses highly sophisticated databases, why not simply build a faster centralized database?

This is exactly the question students should ask.

Blockchain becomes potentially interesting when **multiple organizations need to coordinate state, assets or transactions but should not depend entirely on one conventional database architecture.**

Student Questions

1. Why might a bank prefer a **permissioned blockchain** rather than a public blockchain such as Bitcoin?
2. What is the difference between transferring conventional information about €2 million and transferring a tokenized representation of value?
3. Could a smart contract automatically release payment after an IoT device confirms delivery of the machine? What could go wrong?
4. If banks already have secure centralized databases, what additional problem must blockchain solve to justify its cost and complexity?